Econ 307: Introduction to Econometrics

Bao Linh Hoang

GPA Project

12/14/2025

How to Spend Your Time to Get a Higher GPA?

I. Before Estimation

Time at Denison is just really strange to me. As a student with ADHD, I've perfected the art of procrastination, as cramming one or two nights before an exam and somehow passing. During finals week, my time allocation gets extreme: I sacrifice sleep, meals, and social life to study my life away, while at other times I sleep all day long or hang out instead of using time to study. This pattern raises a question: does how students allocate their time really affect academic performance, and does a perfect student who just spends all day studying have a better GPA?

Research suggests time and preparation do matter. Fu et al. (2025) found that effective time management improves engagement and self-control while reducing distractions such as phone use, while Grave (2011) showed that time devoted to self-study generally raises grades, though excessive academic activity can have diminishing returns. Building on this, my first theoretical model will examine how study hours (HRSTD variable) and extracurricular involvement (HREXTRA variable) relate to the college GPA (COLLGPA).

To capture baseline college students'academic ability, I think it's fair to include two key control variables. The first one is the SAT variable, as SAT scores have long been used as an indicator of cognitive ability and readiness for college-level work. Students with higher SATs tend to perform better academically because they enter with stronger analytical and verbal skills (Noftle & Robins, 2007). The second variable I will use will be the high-school GPA (HSGPA), as it serves as a measure of past academic behavior and consistency. Normally, students who performed well in high school generally have good studying strategies that will help them adapt better to college coursework. Warren and Goins (2019) confirmed that high-school GPA, along with advanced pre-college coursework's scores, are strong predictors of first-semester college success.

Together, these insights motivate the model below, which examines the relationship between standardized test performance, prior high school achievement, and how students allocate their study time.

Model 1:

$$COLLGPA_i = \beta_0 + \beta_1 SAT_i + \beta_2 HSGPA_i + \beta_3 HRSTD_i + \beta_4 HRSTD2_i + \beta_5 HREXTRA + \varepsilon_i$$

COLLGPA: Overall college grade point average

SAT: Aggregate math and critical reading SAT score - out of a maximum score of 1600 (note for those with only ACT scores multiply by 44.5)

- Expected sign: Positive
- SAT scores reflect students' academic preparedness and cognitive ability before college. Noftle and Robins (2007) found that higher SAT scores are associated with traits like conscientiousness, self-discipline, and academic motivation, which surely result in stronger college performance. The students that have higher SAT can demonstrate the characteristics or traits and habits that support sustained academic success. Therefore, I hypothesize that SAT scores have a positive effect on college GPA.

HSGPA: Overall high school grade point average (leave as high school calculated)

- Expected sign: Positive
- High school GPA represents a student's sustained academic performance and ability to handle rigorous coursework. Warren and Goins (2019) found that students who performed well in advanced or honors courses in high school were more successful in college, showing that HSGPA reflects both achievement and readiness for college-level work. Therefore, we expect high school GPA to positively affect college GPA.

HRSTD: Average number of hours spent studying per week (0-168 hours)

- Expected sign: Positive
- Study time represents students' direct investment in academic work. Nonis and Hudson (2010) found that students who devote more time to studying tend to perform better academically, as increased study time enhances understanding and retention of course material. Therefore, we expect study hours to have a positive effect on college GPA.

HRSTD: Average number of hours spent studying per week (0-168 hours) squared

- Expected sign: Negative
- While moderate study time improves academic performance, excessive studying may lead to fatigue, stress, or inefficient learning. Nonis and Hudson (2010) found that study quality matters as much as study quantity. After we pass beyond a certain point, additional hours yield smaller

gains or even negative effects on performance. Therefore, we expect the coefficient for the squared term will be negative, reflecting that the relationship between study time and GPA follows an inverted U-shape.

HREXTRA: Average number of hours spent in extracurricular activities per week (clubs, meeting, job, athletics, community service, etc…)

- Expected sign: Positive
- Participation in extracurricular activities can enhance students' academic success by developing time-management skills, motivation, and a sense of engagement. Bakoban and Aljarallah (2015) found that students who participated in extracurricular activities had significantly higher GPAs than non-participants, suggesting that involvement outside the classroom complements academic achievement. Therefore, we expect hours spent in extracurricular activities to have a positive effect on college GPA.

II. Estimation

Estimation of Theoretical Model 1 Equation:

$$COLLGPA_i = 1.846 + 0.00042SAT_i + 0.2446HSGPA_i + 0.0039HRSTD_i - 0.000206HRSTD2_i - 0.00287HREXTRA_i$$

III. Discussion of Results

Through the regression generated from STATA, now we can take a closer look into the relationship between college GPA and five key independent variables: SAT scores, high school GPA, study hours, the square of study hours, and extracurricular hours.

1. SAT: The coefficient for SAT is 0.000419, indicating that a one-point increase in SAT scores corresponds to a 0.00042 increase in college GPA, holding all other variables constant. This variable is statistically significant at the 5% level ($p = 0.021$), suggesting that standardized test performance remains a meaningful predictor of college success. The result supports prior research showing that SAT scores capture both cognitive ability and conscientiousness (Noftle & Robins, 2007), which contribute to higher academic performance.
2. HSGPA: High school GPA shows the strongest relationship with college GPA, with a coefficient of 0.2446 that is highly significant at the 1% level ($p < 0.001$). This implies that a one-point increase in high school GPA is associated with a 0.245-point increase in college GPA, holding all else constant. The result aligns with Warren and Goins (2019), who found that prior academic achievement and course rigor are strong indicators of college readiness and early success.
3. HRSTD: The coefficient for average study hours per week is 0.0039, but it is not statistically significant ($p = 0.433$). Although the sign is positive, indicating that more study time is associated

with slightly higher GPA, the lack of significance suggests that the quality of study time may matter more than the quantity. This finding echoes Nonis and Hudson (2010), who noted that merely increasing study time does not always translate to better academic performance if students use ineffective study strategies.

4. HRSTD2: The coefficient for the squared term is -0.0000206, which is also not statistically significant ($p = 0.752$). This negative sign reflects the expected diminishing returns to studying, meaning that excessive studying could eventually reduce productivity or increase fatigue. However, since the result is not significant, there is no strong evidence of this nonlinear relationship in the sample.
5. HREXTRA: The coefficient for extracurricular hours is -0.0029, and it is not statistically significant ($p = 0.415$). This suggests that time spent in extracurricular activities does not have a measurable direct effect on GPA in this dataset. Even though we expect the sign to be positive, the relationship between college GPA and time spent on extracurricular activities is negative here. This suggests that there might be variation in how students spend this time that might obscure any consistent academic effect here, as students too focus on out-of-class activities can take away their precious time of doing homework or revisiting their lecture notes.

The model's R^2 value of 0.1584 indicates that approximately 15.8% of the variation in college GPA is explained by the independent variables, while the adjusted R^2 of 0.1307 accounts for model complexity. The F-statistic ($F(5,152)=5.72$, $p < 0.001$), confirms that the model is statistically significant overall, meaning that at least one predictor contributes to explaining GPA. Although this model has not explained a lot yet, this is pretty normal for educational data, which often involve factors that we cannot see such as motivation, teaching quality, and mental health. The F-statistics also allow us to reject the null hypothesis that the independent variables together have no effect on GPA. The VIF test showed no multicollinearity, suggesting that the estimated coefficients are stable and reliable.

From the regression I generated, I think we can conclude that academic preparedness (SAT and HSGPA) plays a more consistent and significant role in predicting GPA than time allocation variables like study or extracurricular hours. From the first model alone, we can think that prior academic foundation is the more reliable predictor of GPA how students use their time.

While we already found that HSGPA and SAT are strong predictors of college GPA in our first model, it does not satisfy the question that I set out at the start of this project: does how students spend their time throughout the day truly affect academic performance? The insignificant results for study and extracurricular hours in Model 1 suggest that total hours alone may not tell the full story. Academic performance may also depend on how time is allocated between activities such as sleep or time spent seeking help. Because the squared term for study hours (HRSTD^2) was not statistically significant (p =

0.752), there was no clear evidence of diminishing returns to studying. Thus we can drop it for now from later models to keep our regression more efficient.

IV. Model Modifications and Discussions

Recent research shows that sleep quality significantly influences learning, memory, and GPA (Hershner & O'Brien, 2023; Curcio et al., 2006). Insufficient or irregular sleep impairs focus and cognitive functioning, so they suggested that well-rested students may perform better even with fewer study hours. Similarly, when you spend more time with tutors, professors, or study sessions, it can be additional guided study time to add on to our previous HRSTD variable, a good indication of time spent on GPA. Studies find that students who seek academic support perform better, as such help strengthens understanding and persistence (Yang, F., & Stefaniak, 2023). Therefore, in Model 2, I include the SLEEP and HELP variables to better capture how students' time devoted to getting enough sleep and guided learning outside would impact on their academic success in Denison.

Model 2:

$$COLLGPA_i = \beta_0 + \beta_1 SAT_i + \beta_2 HSGPA_i + \beta_3 HRSTD_i + \beta_4 HREXTRA + \beta_5 SLEEP_i + \beta_6 STDSLEEP_i + \beta_7 STDEXTRA_i + \beta_7 HELP_i + \varepsilon_i$$

SLEEP: Average hours sleep per week (0-168 hours)

- Expected sign: positive
- Sleep plays a crucial role in maintaining focus, memory, and mental health, all of which affect academic performance. Research shows that adequate sleep supports learning and cognitive function, while sleep deprivation leads to poorer grades (Curcio et al., 2006; Hershner & O'Brien, 2023). Therefore, we expect the coefficient for SLEEP to be positive, as students who get sufficient rest tend to achieve higher GPAs.

STDSLEEP: Interaction terms between hours of studying (HRSTD) and hours of sleep (SLEEP)

- Expected sign: Negative
- This term captures how the relationship between study time and GPA changes depending on sleep. The expectation is that excessive studying combined with reduced sleep leads to fatigue, stress, and less efficient learning. Nonis and Hudson (2010) found that study quality matters as much as study quantity. Beyond a certain point, additional study hours without proper rest may hurt performance. Hence, we expect that the coefficient for this interaction term is expected to be negative.

STDEXTRA: Interaction between hours of studying (HRSTD) and extracurricular hours (HREXTRA)

- Expected sign: Negative
- This variable examines whether balancing study time with extracurricular involvement affects academic performance. While moderate extracurricular engagement supports well-being and time management, too much activity can detract from study time. Therefore, we expect the coefficient for the interaction between study and extracurricular hours to be negative, suggesting that excessive multitasking may lower academic performance. (Bakoban & Aljarallah, 2015)

HELP: Average number of hours per week obtained help outside of class (tutors, professors, study sessions)

- Expected sign: Positive
- Seeking academic help is a productive use of time that improves comprehension and persistence. Studies show that help-seeking is a self-regulated learning behavior positively related to academic outcomes (Yang & Stefaniak, 2023). Thus, we expect the coefficient for HELP to be positive, indicating that guided support boosts GPA.

Estimation of Theoretical Model 2 Equation:

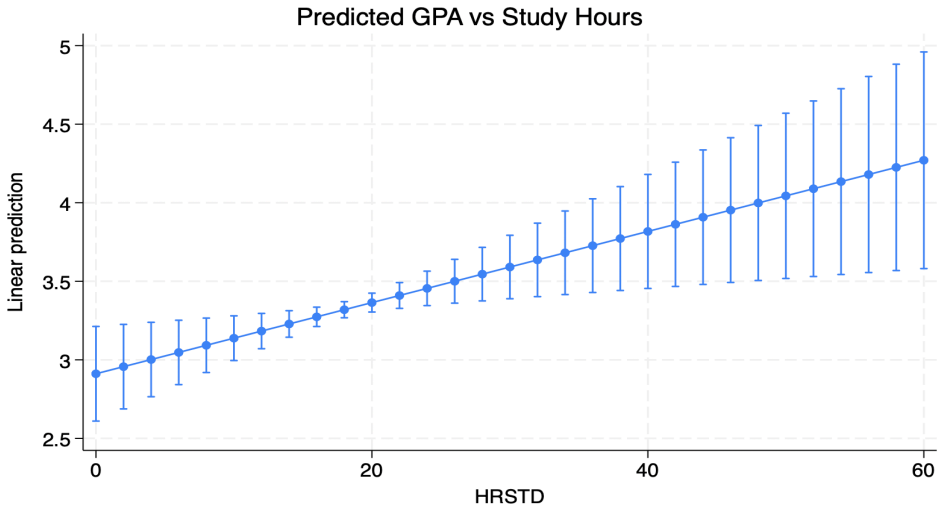
$$COLLGPA_i = 1.84 + 0.00039SAT_i + 0.199HSGPA_i + 0.02HRSTD_i - 0.0045HREXTRA + 0.005SLEEP_i - 0.00034STDSLEEP_i + 0.000136STDEXTRA_i - 0.043HELP_i$$

In this model 2 I've added new time-use variables such as sleep, interaction term between study and sleep, interaction between study and extracurricular hours, and help-seeking hours. Compared to Model 1, this model better explains how students' day to day routine affects their college GPA. The R^2 increases from 0.158 to 0.224, which means the new variables help explain more variation in GPA.

The coefficient for SLEEP is positive but not significant ($\beta = 0.0057$, $p = 0.173$). This shows that getting more sleep may slightly improve GPA, but the effect is somewhat small. The STDSLEEP variable, which measures the interaction between study hours and sleep, is negative and significant ($\beta = -0.00034$, $p = 0.026$). This means that students who study a lot but do not sleep enough tend to have lower GPA, so I think time is a zero sum game here. It suggests that lack of rest can reduce the benefit of studying. The STDEXTRA term is positive but insignificant. This means that extracurricular activities do not have a clear relationship with GPA when other factors are included. The one variable I find interesting here is HELP. It is both negative and significant ($\beta = -0.043$, $p = 0.004$). Although this seems surprising at first, if you think about it, it may be because students who spend more time seeking help and going to the office and TA hours are the folks who are already struggling in their classes. SAT and high school GPA remain

strong and significant predictors, and study hours become more meaningful if we add more time-balance factors into the model.

From this I conclude that Model 2 improves on Model 1 by showing that how students manage their study time, sleep, and academic support matters for performance. The results suggest that a balance is really important, as studying longer without enough sleep might not equal better grades.



To test the relationship between COLLGPA and HRSTD, I have generated a graph here. We can see that there is a clear positive relationship between college GPA and study hours. As students spend more time studying, their predicted GPA generally increases, which is predictable. However, the error bars widen when we move past the 20 hours per week mark, meaning there is great variability in outcomes among heavy studiers. This suggests that while studying more often correlates with higher GPA, excessive study time does not guarantee improvement. Some students may experience diminishing returns. Things like fatigue, stress, or inefficient study habits can happen if they study too much. So, being a "perfect" student who studies nonstop doesn't necessarily yield the best results; effective time management is key here.

However, there's still more room for improvement, as STDEXTRA is still pretty statistically insignificant, suggesting that the interaction between studying time and extracurricular activities does not contribute much to GPA; hence, we should drop it. In addition, we can add more variables in this equation to bump up R^2 to be more inclusive to other facets of a students' lives, as their lives do not only evolve around studying, sleep, and extracurricular activities.

For these reasons, in Model 3, I want to focus on personal characteristics, specifically, whether a student is an athlete, a member of Greek life, or has an academic disability, which can lowkey shape how time is used and influence academic outcomes. I'm particularly interested in these three variables because I've seen firsthand how Denison football players juggle demanding practice schedules, how my friends in

sororities and fraternities balance constant meetings and philanthropy events, and how, as someone with ADHD, I personally struggle to stay focused without medication and often lose time to distraction.

Model 3:

$$COLLGPA_i = \beta_0 + \beta_1 SAT_i + \beta_2 HSGPA_i + \beta_3 HRSTD_i + \beta_4 HREXTRA + \beta_5 SLEEP_i + \beta_6 STDSLEEP_i + \beta_7 Disability_i + \beta_8 ATHLETE_i + \beta_9 GREEK_i + \varepsilon_i$$

Disability: Dummy variable on whether a student have an academic disability or not (where 1 = Yes, and 0 = No)

- Expected sign: Negative
- Students with academic disabilities, such as ADHD or other learning disorders, often face additional challenges in sustaining attention, managing coursework, and adjusting to academic expectations. Even with accommodations, these factors can make achieving high grades more difficult. Research shows that college students with ADHD or similar conditions earn significantly lower first-year GPAs than their peers without such diagnoses (Gormley, DuPaul, Weyandt, & Anastopoulos, 2016). Consequently, we expect the disability variable to have a negative relationship with college GPA.

ATHLETE: Dummy variable on whether a student is an intercollegiate Athlete or not (where 1 = Yes, and 0 = No)

- Expected sign: Negative
- Participation in intercollegiate athletics often demands a lot of time for a school athlete for training, practices, travel, and competitions. These kinds of commitments can reduce the time available for students to spend on studying, rest, and other academic work (Parham, 1993). Therefore, we expect being an athlete to be associated with a lower college GPA.

GREEK: Dummy variable on whether a student have greek affiliation or not (where 1 = Yes, and 0 = No)

- Expected sign: Negative
- Involvement in Greek life often requires time for social events, recruitment, and organizational duties, which can conflict with academic responsibilities. Empirical research consistently finds that fraternity and sorority membership is associated with lower GPAs, roughly a 0.1-0.3 decrease on a 4.0 scale, after controlling for academic background (Davis, Mara, & Schmidt, 2018). These obligations may reduce study time and increase stress or distraction, so we expect Greek affiliation to negatively impact college GPA.

Estimation of Theoretical Model 3 Equation:

$$COLLGPA_i = 1.6 + 0.00039\ SAT_i + 0.194HSGPA_i + 0.022HRSTD_i - 0.016HREXTRA + 0.007SLEEP_i - 0.00036STDSLEEP_i + 0.1409Disability_i - 0.031ATHLETE_i + 0.214GREEK_i$$

I built model 3 on the previous model 2 but now I have 3 new dummy variables: Disability, ATHLETE, and GREEK, so I can capture time commitments and disorder challenges that might influence GPA. The model's explanatory power improves compared to earlier models. We have R^2 of 0.2595 and an adjusted R^2 of 0.2145, which in simple language means that the included variables explain about 21% of the variation in college GPA, which is way better than 18% in Model 2 and 13% in Model 1. The overall F-statistic ($F = 5.76$, $p < 0.001$) confirms that the model is statistically significant as a whole. Now let's take a look at the new dummy variables.

The coefficient for Disability ($\beta = -0.141$, $p = 0.073$) is negative, and only marginally significant. Therefore, we can see that students with academic disabilities tend to have slightly lower GPAs. The findings from Gormley et al. (2016) agrees with this: ADHD and similar learning challenges can still reduce academic consistency even if we have support services. Interestingly, the ATHLETE dummy's coefficient is also negative ($\beta = -0.031$, $p = 0.621$) and not statistically significant, so being an intercollegiate athlete may not directly affect GPA once study habits and other factors are controlled for. Therefore, we can see that Denison is doing a pretty good job at supporting our student-athletes academically , helping them balance athletic and academic commitments. Another interesting observation is that Greek affiliation has a highly positive and highly significant effect ($\beta = 0.214$, $p < 0.001$), which is opposite to my initial hypothesis that Greek life would lower GPA. I think that having academic networking and social support within fraternities and sororities, can help study cooperation even if they have heavy social demands. My friends at the sororities have been on the Dean's List for the 4th time consecutively now, so that checks out.

In essence, this new model that we have provides a more nuanced understanding of time allocation and clubs/ Greek commitments. While studying and sleep continue to matter most, structured identities like Greek membership or having a disability also influence GPA.

Seeing how weak the association of ATHLETE is, I think we should do a hypothesis test for this variable.

1. Set up

   H_0: $\beta_8 = 0$

   H_a: $\beta_8 \neq 0$
2. Find the critical t-value

   $n = 158$, $k = 9$

   $df = 148$

   Significance level (α): two-tailed at 0.05

   $\rightarrow t_{(c)} \approx 1.976$

3. Calculate the t-value

   $\beta_8 = -0.0312$, SE = 0.0629

   $t(R) = -0.0312 / 0.0629 = -0.50$

4. Decision rule

   If $|t(R)| > t(c) \rightarrow$ Reject H_0

   If $|t(R)| \leq t(c) \rightarrow$ Fail to reject H_0

As we can see, 0.50 is smaller than 1.976, so we fail to reject H_0. Therefore, athletic participation does not significantly affect GPA at the 5% level. I think it's best that we leave this variable for the next model.

For this reason, we could drop the dummy variable ATHLETE here. In addition, let's look at another major part of student life: social behavior and alcohol consumption, as a lot of the variables in the previous that focus on the time allocation aspect of Denisonians have lower statistical significance (by adding more time-related variables, we can improve the model predictive power). I've seriously seen how my friends spend nights drinking, pregaming, or recovering from hangovers, which take their study time or sleep away. Therefore I want to test this time-allocation tradeoff with 2 new variables, Mid_drinker, Big_drinker and SLEEP2, to see if heavier drinking negatively impacts GPA compared to students who do not or drink very little.

Model 4:

$$COLLGPA_i = \beta_0 + \beta_1 SAT_i + \beta_2 HSGPA_i + \beta_3 HRSTD_i + \beta_4 HREXTRA + \beta_5 SLEEP_i + \beta_6 SLEEP2_i + \beta_7 STDSLEEP_i + \beta_8 Disability_i + \beta_9 GREEK_i + \beta_{10} Mid_drinker_i + \beta_{11} Big_drinker_i + \varepsilon_i$$

SLEEP2: Average hours of sleep per week (squared)

- Expected sign: Negative
- $SLEEP^2$ captures the nonlinear (diminishing) effect of sleep on GPA. While getting enough sleep improves cognitive functioning, memory consolidation, and focus, having too much sleep can cause fatigue, stress, or poor time management. Studies show an optimal range of sleep enhances academic outcomes, but too much or too little sleep can reduce performance (Gilbert & Weaver, 2010). Therefore, SLEEP2 tests this inverted U-shaped relationship between sleep and GPA.

Mid_Drinker: Dummy variable for moderate alcohol consumption (DRINK == 8 and 15 - reference group = non-drinkers or those who drink very little)

- Expected sign: Negative
- For this categorical term, I'll let alcohol consumption between 1 and 14 drinks per week moderate. Compared to light social drinking with no effects, even moderate consumption can reduce concentration, study hours, and sleep quality. Thus, I expected the sign to be slightly

negative, but small positive effects are possible if drinking reflects social engagement that supports wellbeing.

Big_Drinker: Dummy variable for moderate alcohol consumption (DRINK == 22 and 29+ - reference group = non-drinkers or those who drink very little)

- Expected sign: Negative
- Heavy drinking is consistently linked to lower GPAs and increased class absence, poor time management, and lower brain performance (Paschall & Freisthler, 2003). Heavy drunk students can experience academic and behavioral consequences that interfere with learning. Therefore, I expect a negative coefficient for big drinkers, because it is detrimental to academic achievement.

Estimation of Theoretical Model 4:

$$COLLGPA_i = 2.22 + 0.00027SAT_i + 0.183HSGPA_i - 0.025HRSTD_i - 0.001HREXTRA - 0.0148SLEEP_i + 0.0002SLEEP2_i - 0.0004STDSLEEP_i + 0.128Disability_i + 0.2121GREEK_i - 0.0117Mid_drinker_i - 0.132Big_drinker_i$$

With these 3 new variables, the model's overall fit improves, with R^2 increasing from 0.2595 in Model 3 to 0.2923, so it's safe to say that these new variables add modest explanatory power. Here's a cool thing: SLEEP now has a negative coefficient when we add SLEEP2 (which is positive) when we turn a linear relationship to a curved one. The significant level ($p = 0.008$) indicates a U-shaped relationship: both too little and too much sleep may be associated with lower GPA, while moderate, balanced sleep supports better performance. Okano's studies agree, which found that consistent, adequate sleep improves cognitive functioning and academic results. Unsurprisingly for drinking behavior, Big_drinker has a negative coefficient (-0.1379) and is marginally significant ($p \approx 0.10$), supporting past findings (Paschall & Freisthler, 2003) that heavier alcohol use tends to harm academic performance. Drinking just moderately does not matter to GPA compared to non-drinkers, as Mid_drinker are pretty low on statistical significance. I think Model 4 helps us understand more about how time and lifestyle impact academic performance. Of course, study time and consistent sleep patterns promote higher GPA. Heavy drinking can offset these gains, so we should care more about time use and drinking habits all together.

As we can see here, the explanatory power of Model 4 has improved a lot from our first Model 1, in terms of R^2, as it 32.3% of the variables in this model. By adding drinking habit of college students, whose GPA is more likely to be negatively impacted in terms of sleep and studying time, we can have more holistic pictures of how time allocation can impact academic performance. Now, in our last model, we just need to add one more log term for HREXTRA from Model 4, as the linear relationship alone of our last important variable, HREXTRA, has not satisfied much statistically in our COLLGPA regression model.

Model 5:

$$COLLGPA_i = \beta_0 + \beta_1 SAT_i + \beta_2 HSGPA_i + \beta_3 HRSTD_i + \beta_4 HREXTRA + \beta_5 SLEEP_i + \beta_6 SLEEP2_i + \beta_7 STDSLEEP_i + \beta_8 Disability_i + \beta_9 GREEK_i + \beta_{10} Mid_drinker_i + \beta_{11} Big_drinker_i + \beta_{12} ln_HREXTRA + \varepsilon_i$$

ln_HREXTRA: Natural log of extracurricular hours per week

- Expected sign: Positive
- Coming back to extracurricular activities, we can use a log-term to see if the HREXTRA improves skills like time management and motivation, which can boost GPA. Of course we only have 24 hours everyday, too much participation can reduce study time and our focus and hurt GPA. Taking the natural log captures these diminishing returns, where early involvement helps but excessive hours add less benefit (Eccles et al., 2003; Bakoban & Aljarallah, 2015).

Estimation of Theoretical Model 5 Equation:

$$COLLGPA_i = 2.51 + 0.0002SAT_i + 0.18HSGPA_i + 0.024HRSTD_i + 0.0128HREXTRA - 0.0163SLEEP_i + 0.00027SLEEP2_i - 0.00041STDSLEEP_i - 0.147Disability_i + 0.2399GREEK_i - 0.022\ Mid_drinker_i - 0.145Big_drinker_i - 0.162 ln_HREXTRA$$

In this model 5, I decided to use ln_HREXTRA, the logarithmic transformation of extracurricular hours, to capture the diminishing returns of time spent in for extracurricular activities. Interestingly, opposite to our first guess, the ln_HREXTRA variable turns slightly negative (−0.161, p = 0.112) while HREXTRA coefficient turns positive from Model 4 (same case happen for our SLEEP variable), suggesting that after a certain threshold, the time we use on extracurricular activities may detract from academic focus.

Compared to the last model, overall model fit does improve again: R^2 rises to 0.3235 (from 0.2923). So putting an extra non-linear extracurricular effect is a good choice, as it enhances explanatory power. While HSGPA is still a huge contributor in predicting COLLGPA, more variables/ characteristics of Denisonians has now shown to be reliable indicators of the model including Disability, GREEK, Big_drinker, and ln_HREXTRA, which has satisfied the goal that we set out since the start of the project of understanding how habit and time allocation of Denisonians impact their college performances.

V. Post Estimation

A. Best model

Now I think we can stop at Model 5, since it's the best-performing specification in this analysis because it combines academic background, time-allocation variables, and personal or lifestyle factors that we can explain comprehensively about the variations in college GPA. Compared to earlier models, Model 5

yields the highest R-squared value (0.3235) and the lowest Root MSE (0.3228), meaning it explains the largest share of GPA variation with the smallest prediction error. When we have behavioral variables (like HRSTD, HREXTRA, and SLEEP) and structural characteristics (like Disability, GREEK and DRINK), Model 5 captures a more holistic view of the nature of student performance at Denison, in association to how they spend time at school. It also improves upon the latest Model 4 by adding the log of extracurricular hours (ln_HREXTRA), which accounts for diminishing returns from excessive involvement outside academics, showing that balance, not just effort, drives success.

When we look at our predictions, Model 5 maintains the expected signs for most coefficients and meaningful significance for core variables such as HSGPA, HRSTD, GREEK, and the non-linear SLEEP2 term. These results indicate that academic preparation, study habits, and balanced rest remain key predictors of academic performance, while other factors like excessive drinking or having academic disabilities have consistent negative associations.

I now firmly believe that Model 5 is the most complete and statistically reliable model that I have produced so far because it integrates cognitive, behavioral, and lifestyle dimensions, providing a fuller picture of what shapes Denison student GPA outcomes in regards to the theme of time that I want to dive into when I start on this project.

B. Test the hypothesis

1. Set up

   $H_0 : \beta_1 \leq 0$

   $H_A : \beta_1 > 0$

2. Find the critical t-value

   (n = 157, k = 12)

   Degrees of freedom $(df) = n - k - 1$ = 157 - 12 - 1 = 144

   Significance Level (α): A one-tailed test at $\alpha = 0.05$

   → Critical t-value (t_c) is approximately 1.645.

3. Calculate the t-value

   $\beta_1 = 0.0002326$, standard error (SE) = 0.0001727

   → t-value (t_r) = 0.0002326 / 0.0001727 = 1.35

4. Decision rule

   If $|t_R| > t_c$ → Reject H_0

   If $|t_R| \leq t_c$ → Fail to Reject H_0

Because 1.35 < 1.645, we Fail to Reject H_0. Therefore, an individual's combined SAT score does not significantly increase college GPA at the 5% level. However, with a one-sided p-value of 0.09, the result

is marginally significant at the 10% level, suggesting a weak positive association between SAT and GPA when accounting for additional behavioral and lifestyle variables.

C. Does SAT have the greatest impact on college GPA?

No, SAT does not have the greatest impact on college GPA in Model 5. Although the SAT coefficient (0.0002326) is positive, its t-value (1.35) and p-value (0.18) show that it is not statistically significant at the 5% level. In contrast, high school GPA (HSGPA) and hours spent studying (HRSTD) have larger and statistically significant effects on GPA ($p = 0.002$ and $p = 0.005$, respectively). This indicates that prior academic performance and study habits are stronger predictors of college success than standardized test scores once other behavioral and lifestyle factors are controlled for. I mean I, for example, had a perfect SAT score, but I am still struggling everyday to keep up my grades as my schedules tend to be messy and I do not take care of myself well, which has been taking a toll on my grades generally. Hence, in my opinion, this model 5 captures my college life story pretty perfectly.

D. One-sentence interpretations for each independent variable in the best model:

- SAT: Holding other factors constant, a one-point increase in SAT score is associated with a 0.0002-point rise in college GPA, meaning a 100-point SAT increase predicts roughly a 0.02-point higher GPA.
- HSGPA: Holding other factors constant, each one-point increase in high school GPA corresponds to a 0.18-point increase in college GPA, confirming that prior academic performance strongly predicts college success.
- HRSTD (Study Hours): At the mean study time, every additional hour spent studying per week raises college GPA by 0.024 points, indicating that more consistent study time tends to improve academic outcomes, holding other factors constant,
- HREXTRA (Extracurricular Hours): Each extra hour devoted weekly to extracurricular activities is linked to a 0.0128-point GPA increase, suggesting moderate involvement may enhance academic balance and engagement, holding other factors constant,
- SLEEP: Holding other factors constant, each additional hour of sleep per week is associated with a 0.0163-point decrease in GPA, but this negative effect reverses at moderate levels once the squared term is considered.
- SLEEP2: The positive coefficient of 0.00027 confirms a U-shaped relationship, after about 30 hours of weekly sleep (about 4 to 5 hours per night), additional sleep begins to improve GPA, holding other variables constant.

- STDSLEEP: Students with a registered disability have, on average, a 0.147-point lower GPA than those without, holding all other factors constant.
- Disability: Students with a registered disability have, on average, a 0.147-point lower GPA than those without, holding all other factors constant, holding other variables constant.
- GREEK: Students in fraternity or sorority are associated with a 0.24-point higher GPA than those who are not, holding other variables constant.
- Mid_drinker: Moderate drinkers have GPAs about 0.02 points lower than non-drinkers or light drinkers, though the effect is small and statistically insignificant, holding other variables constant.
- Big_drinker: Heavy drinkers show an average 0.15-point lower GPA compared to non- or light-drinkers, holding other variables constant.
- ln_HREXTRA: Taking the natural log of extracurricular hours, a one-percent increase in activity time predicts roughly a 0.0016-point decrease in GPA, holding other variables constant.

E. My GPA prediction using our best model

2.51 + 0.0002*(1600) + 0.18*(3.8) +0.024*(35) +0.0128*(3) -0.0163*(35) + 0.00027*(35^2) - 0.00041*(35*35) - 0.147*(1) + 0.2399*(0) - 0.022*(0) - 0.145*(0) - 0.162*ln(3) = 3.325

My actual GPA is 3.6, so there is a 3.6 - 3.325 or 0.275 point difference between my model and the actual GPA. I believe that our final model 5 is a pretty solid model that does a pretty decent job computing the College GPA and capturing a holistic picture of my college life and hopefully other Denisonians as well.

VI. Professor Juraqulova's Model

$$COLLGPA_i = \beta_0 + \beta_1 SAT_i + \beta_2 ln_HSGPA_i + \beta_3 HRSTD_i + \beta_4 HRSTD2_i + \beta_5 DECISION_i + \beta_6 FEM_i + \beta_7 SAT_FEM_i + \beta_8 ECON_i + \varepsilon_i$$

From codebook:

- COLLGPA: Overall college grade point average
- SAT: Overall SAT score
- ln_HSGPA: The natural logarithm of the overall high school grade point average
- HRSTD: Average number of hours spent studying per week (0-168 hours)
- HRSTD2: Average number of hours spent studying per week (0-168 hours) square
- DECISION: Apply to Denison using Early Decision (1=Yes/0=No)
- FEM: Gender (M=0/F=1)
- SAT_FEM: Interaction term between Female and SAT
- ECON: Student who choose Economics major (1=Yes/0 = not Econ major)

Estimation of JURAQULOVA Model:

$$COLLGPA_i = 1.093647 + 0.0004656SAT_i + 1.157828ln_HSGPA_i + .0018735HRSTD_i + 1.79e\text{-}06HRSTD2_i + 0.0029234DECISION_i + 0.247362FEM_i - 0.00009SAT_FEM_i + 0.0133352ECON_i$$

B. Interpretation of JURAQULOVA's model

- SAT: Holding other factors constant, a one-point increase in SAT score is associated with a 0.00047 point increase in college GPA, meaning students with higher SAT scores tend to earn slightly higher GPAs.
- ln_HSGPA: Holding other factors constant, a 1% increase in high school GPA is associated with a 0.0116 point increase in college GPA, which means that strong academic performance in high school predicts higher college success but with diminishing returns at the top.
- HRSTD2: At the mean study time (about 20 hours per week), the marginal effect is calculated as 0.0018735 + 2 (1.79e−06) (20) = 0.0019441, meaning that at average study levels, an additional hour of studying per week is associated with roughly a 0.0019 increase in GPA, holding other factors constant. However, since the squared term is very small and statistically insignificant, the overall effect of additional study hours is positive but modest.

C. Hypothesis Test for JURAQULOVA's model

Using the JURAQULOVA's model, we can now test the hypothesis "An individual's combined math and critical reading SAT score significantly influences his/her college GPA."

1. Set up

   $H_0 : \beta_1 = 0$

   $H_A : \beta_1 \neq 0$
2. Find the critical t-value

   (n = 157, k = 8)

   Degrees of freedom (df) = $n - k - 1$ = 157 - 8 - 1 = 148

   Significance Level (α): A two-tailed test at α= 0.05

   → Critical t-value (t_c) is approximately 1.976.
3. Calculate the t-value

   β_1 =.0004656 , standard error (SE) = 0.0002179

   → t-value (t_R)= .0004656/.0002179 = 2.137
4. Decision rule

   If $|t_R| > t_c$ → Reject H_0

   If $|t_R| \leq t_c$ → Fail to Reject H_0

Because 2.137 > 1.976, we reject H_0. Therefore, an individual's combined math and reading SAT score significantly influences his/her college GPA, at the 5% level. The positive coefficient implies that higher SAT scores are associated with higher GPAs.

D. JURAQULOVA Model vs Model 5

Comparing the JURAQULOVA model and Model 5, I think that Model 5 provides a better explanation of college GPA. Firstly, we can see that the JURAQULOVA model, which included eight predictors, achieved an R^2 of 0.1953 and an adjusted R^2 of 0.1518. However, Model 5, with twelve predictors, raised these values to 0.3235 and 0.2671 respectively. This means that we have 11 percent more of the variation in GPA even after accounting for model complexity. The overall F-statistic also increased from 4.49 ($p = 0.0001$) to 5.74 ($p < 0.001$), confirming that model 5 is stronger overall in statistical significance. In the JURAQULOVA model, SAT and high-school GPA were the only significant predictors, whereas in Model 5, additional variables such as study hours (HRSTD), sleep patterns, and Greek organization membership became important. In my model 5, the SAT coefficient lost significance once I included more lifestyle and behavioral factors. It's safe to say that academic performance in college depends not only on cognitive measures like test scores but also on personal habits and social engagement of the students. Overall, I do believe that Model 5 is more comprehensive and provides a more accurate representation of the factors influencing college GPA. However, we also should understand that the models I built revolved around time allocation and habits so they catered more toward students that are heavily influenced by their schedules. Therefore, Dr. J's are still a good predictor of GPA for her own variables and worth using (her model even predicted my actual GPA really well).

VII. Conclusion and final recommendation

Honestly, Model 5 makes one thing pretty clear: how we spend our time at Denison actually matters for our grades. Sure, SAT and high school GPA still tell part of the story, but once you throw in real-life habits like how much we study, sleep, and get involved on campus, the picture gets changed fast. Regular studying and being active in Greek life seem to boost GPA, while chaotic sleep schedules or pulling all-nighters definitely don't help. What's funny is that the SAT score kind of loses its power here, suggesting that what you do after getting to college matters more than whatever score you came in with.

From a bigger-picture view, this means schools like Denison shouldn't just focus on test prep or tutoring. Helping students find a balance, like learning to manage time, get enough sleep, and stay involved without burning out, might actually do more for academic success. Honestly, that balance is something I still struggle with (hello, ADHD and caffeine-fueled cramming), but this model really drives home that success isn't about doing everything and being a perfect student, it's about managing your time well enough to do the important things right.

Summary of Regression Results: College GPA and Explanatory Variables						
	M1	M2	M3	M4	M5	JURAQULOVA
SAT	0.000419*	0.000392*	0.000395*	0.000272	0.000232	0.000465*
	(2.34)	(2.26)	(2.26)	(1.58)	(1.35)	(2.14)
HRSTD	0.004	0.020*	0.022**	0.025**	0.024**	0.002
	(0.79)	(2.28)	(2.67)	(2.99)	(2.88)	(0.37)
HSGPA	0.245***	0.199***	0.194**	0.183**	0.182**	
	(4.19)	(3.42)	(3.35)	(3.25)	(3.23)	
HREXTRA	-0.003	-0.005	-0.002	-0.001	0.013	
	(-0.82)	(-0.75)	(-0.42)	(-0.31)	(1.43)	
SLEEP		0.006	0.007	-0.015	-0.016	
		(1.37)	(1.73)	(-1.65)	(-1.81)	
STDSLEEP		-0.000343*	-0.00036*	-0.0004**	-0.00041**	
		(-2.24)	(-2.41)	(-2.83)	(-2.72)	
Disability			-0.141	-0.128	-0.147	
			(-1.81)	(-1.69)	(-1.90)	
GREEK			0.214***	0.212***	0.240***	
			(3.85)	(3.85)	(4.24)	
Mid_drinker				-0.012	-0.022	
				(-0.20)	(-0.37)	
Big_drinker				-0.133	-0.145	
				(-1.60)	(-1.75)	
SLEEP2				0.002**	0.00027**	
				(2.62)	(2.76)	
ln_EXTRA					-0.162	
					(-1.60)	

HRSTD2	-0.000021					1.79e-06
	(-0.32)					(0.03)
ln_hsgpa						1.158***
						(4.16)
DECISION						0.003
						(0.05)
FEM						0.247
						(0.5)
SAT_Fem						−0.00009
						(-0.24)
ECON						0.013
						(0.21)
STDEXTRA		0.000136				
		(0.47)				
HELP		-0.043**				
		(-2.93)				
ATHLETE			-0.031			
			(-0.50)			
Constant	1.847***	1.842***	1.601***	2.222***	2.507***	1.094*
	(6.06)	(5.05)	(4.49)	(5.36)	(5.59)	(2.45)
N	158	158	158	158	157	157
R^2	0.158	0.224	0.259	0.306	0.323	0.195
Adjusted R^2	0.1307	0.1823	0.2145	0.2535	0.2671	0.1518
t statistics in parentheses						
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$						

References

Bakoban, R. A., & Aljarallah, S. A. (2015). Extracurricular activities and their effect on the student's grade point average: Statistical study. *Journal of Education and Practice, 6*(11), 129–142.

Bakoban, R. A., & Aljarallah, S. A. (2015). Extracurricular activities and their effect on the student's grade point average: Statistical study. *Educational Research and Reviews, 10*(20), 2737–2744.

Curcio, G., Ferrara, M., & De Gennaro, L. (2006). Sleep loss, learning capacity and academic performance. *Sleep Medicine Reviews, 10*(5), 323–337. https://doi.org/10.1016/j.smrv.2005.11.001

Davis, M., Mara, C., & Schmidt, E. (2018). *Social animal house: The economic and academic consequences of fraternity membership.* Union College Department of Economics Working Paper.

Eccles, J. S., Barber, B. L., Stone, M., & Hunt, J. (2003). Extracurricular activities and adolescent development. *Journal of Social Issues, 59*(4), 865–889.

Fu, Y., Wang, Q., Wang, X., Zhong, H., Chen, J., Fei, H., Yao, Y., Xiao, Y., Li, W., & Li, N. (2025). Unlocking academic success: The impact of time management on college students' study engagement. *BMC Psychology, 13*(1), 323. https://doi.org/10.1186/s40359-025-02619-x

Fugate, M. E. (2023). *Time allocation of undergraduate students: How students spend their time as it relates to academic and engagement outcomes* (Order No. 31508141). *ProQuest Dissertations and Theses Global.* The University of Toledo. Retrieved from https://denison.idm.oclc.org/login?url=https://www.proquest.com/dissertations-theses/time-allocation-undergraduate-students-how-spend/docview/3075635343/se-2

Gilbert, S. P., & Weaver, C. C. (2010). Sleep quality and academic performance in university students: A wake-up call for college psychologists. *Journal of College Student Psychotherapy, 24*(4), 295–306. https://doi.org/10.1080/87568225.2010.509245

Gormley, M. J., DuPaul, G. J., Weyandt, L. L., & Anastopoulos, A. D. (2016). First-year GPA and academic service use among college students with and without ADHD. *Journal of Attention Disorders, 23*(14), 1766–1779. https://doi.org/10.1177/1087054715623046

Grave, B. S. (2011). The effect of student time allocation on academic achievement. *Education Economics, 19*(3), 291–310. https://doi.org/10.1080/09645292.2011.585794

Hershner, S., & O'Brien, L. M. (2023). The impact of sleep on academic performance in college students. *Sleep Health, 9*(2), 125–134. https://pubmed.ncbi.nlm.nih.gov/36780521

Noftle, E. E., & Robins, R. W. (2007). Personality predictors of academic outcomes: Big five correlates of GPA and SAT scores. *Journal of Personality and Social Psychology, 93*(1), 116–130. https://doi.org/10.1037/0022-3514.93.1.116

Nonis, S. A., & Hudson, G. I. (2010). Performance of college students: Impact of study time and study habits. *Journal of Education for Business, 85*(4), 229–238. https://doi.org/10.1080/08832320903449550

Okano, K., Kaczmarzyk, J. R., Dave, N., Gabrieli, J. D. E., & Grossman, J. C. (2019). Sleep quality, duration, and consistency are associated with better academic performance in college students. *NPJ Science of Learning, 4*(16). https://doi.org/10.1038/s41539-019-0055-z

Parham, W. D. (1993). The intercollegiate athlete: A 1990s profile. *The Counseling Psychologist, 21*(3), 411–429. https://doi.org/10.1177/0011000093213005

Paschall, M. J., & Freisthler, B. (2003). Does heavy drinking affect academic performance in college? *Journal of Studies on Alcohol, 64*(5), 515–519.

Paschall, M. J., & Freisthler, B. (2003). Does heavy drinking affect academic performance in college? Findings from a prospective study of high achievers. *Journal of Studies on Alcohol, 64*(4), 515–519. https://doi.org/10.15288/jsa.2003.64.515

Presley, C. A., Meilman, P. W., & Cashin, J. R. (1996). *Alcohol and drugs on American college campuses: Use, consequences, and perceptions of the campus environment.* The Core Institute, Student Health Institute, University of Southern Maine.

Warren, J. M., & Goins, C. L. (2019). Exploring the relationships between high school course enrollment, achievement, and first-semester college GPA. *Journal of Educational Research and Practice, 9*(1), 386–399. https://doi.org/10.5590/JERAP.2019.09.1.27

Wechsler, H., Lee, J. E., Kuo, M., & Lee, H. (2002). Trends in college binge drinking among men and women: Results from four Harvard School of Public Health College Alcohol Study surveys: 1993–2001. *Journal of American College Health, 50*(5), 203–217. https://doi.org/10.1080/07448480209595713

Yang, F., & Stefaniak, J. (2023). A systematic review of studies exploring help-seeking strategies in online learning environments. *Online Learning, 27*(1), 107–126. https://doi.org/10.24059/olj.v27i1.3400